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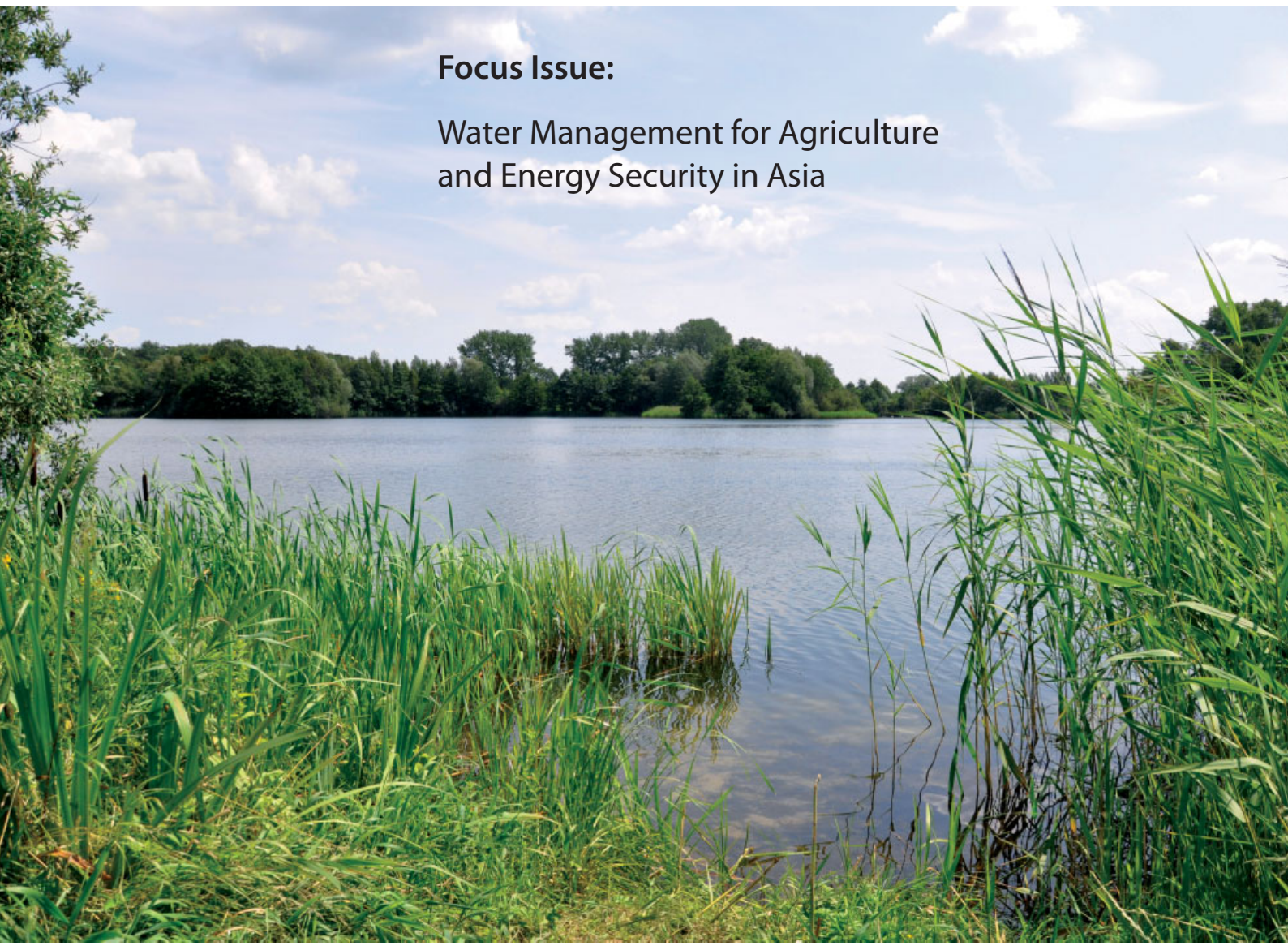
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Research Article

Effectiveness of *Dichanthium annulatum* in Watercourses for Reducing Sediment Delivery from Agricultural Watersheds

A field experiment was conducted to determine the efficacy of *Dichantium annulatum* grass filters in watercourses and to optimize the grass coverage in these watercourses. This experiment was conducted in 2% slope watercourses with varying filter grass coverage (0, 25, 50, 75, 100%) as main treatments and different flow rates (5, 10, 15, 20 L/s) as sub-treatments. The inflow and outflow, flow velocity, depth of the flow, and sediment concentration was measured and analyzed for each run. The experimental results reveal that the grass filter strips are able to reduce the sediment concentration in the runoff water by six times (from 4.2 to 0.65 g/L). The filter strips are able to reduce the velocity of the runoff water by two times (from 1.06 to 0.47 m/s) and convert the supercritical flows into subcritical flows. The filter strips are able to increase the Manning's coefficient from 0.0323 to 0.1006 under various discharges. From the experiment, it can be concluded that more than 50% grass filter coverage in watercourses did not further reduce sediment concentration, outflow, flow velocity, and the Froude number and increase roughness of the channels. Based on these results, it is recommended that *D. annulatum* can be used as a filter in watercourses to prevent runoff and soil losses from crop fields and thereby reducing the sedimentation of downstream water bodies. The optimum grass coverage, i.e. 50% of the channel length, can maintain the maximum grass production with minimum soil loss.

Keywords: Froude number; Grass filter; Manning's coefficient; Sediment concentration

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1 Introduction

The runoff generated from cultivable lands that get merged at different places, becomes concentrated and passes through natural channels or drains at greater velocity. The runoff carrying sediments which have been eroded from crop fields and watercourses, etc., continues to be a major non-point source of pollution for surface water bodies in the entire world. It can affect the stream habitat, the storage capacity of the structures and the quality of stored water in storage structures. Non-point source water pollution of streams and lakes is a major problem in agricultural croplands [1]. Particularly in a monsoonal country like India, high intensity rains during the monsoon season causes severe erosion in croplands which leads to sedimentation of the water bodies. In India, several mechanical or engineering soil and water conservation measures have been implemented by various Governmental Departments like the Watershed Development or the Water Management Agencies and other non-governmental agencies through various integrated watershed development programs/schemes to check soil erosion and development of water resources for the sustainable use in a watershed basis. The mechanical measures such as contour and graded bunds, terraces, check dams, etc., are able to reduce soil erosion and due to that increase the production. However, these structures can minimize the runoff flow to the downstream reservoirs sometimes and the downstream reservoirs will not get the sufficient inflow. The studies [2, 3] reported a reduced runoff flow to the downstream water bodies as a result of water harvesting along drainage lines and within agricultural and forested areas. It is also mentioned that the construction of mechanical measures involves very high cost and farmers are not able to implement these measures without support of any schemes/programs. There is a need to search alternative bio-engineering or vegetative mechanisms to these measures in which cost of construction can be minimized and downstream reservoirs will receive a sediment-filtered clean inflow.

The vegetative soil and water conservation measures such as vegetative filter strips, grassed watercourses, vegetative barriers, grassed buffer strips, and riparian buffer zones adjacent to source areas have been used as controls to reduce on-site effect of erosion and off-site transport of sediments. The live bunds or vegetative barriers are the alternative biological measures for reducing the surface runoff with increasing the infiltration time and effectively

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conserving soil and water [4]. Several studies from India have been reported on vegetative barriers for reducing the sediment loads from crop fields [5–11]. In vegetated buffers, filters are designed to remove sediment and other pollutants from surface runoff by filtration, deposition, and infiltration [12]. The efficacy of vegetated buffers and vegetative barriers in erosion and pollutants control has already been reported by numerous studies and most of the studies showed a positive effect on reducing the transfer of sediments, nutrients, and pesticides to surface water [13–17]. Buffer vegetation, especially grass, acts as a filter by increasing the surface roughness, which augments infiltration by decreasing flow volumes and velocity [18]. The vegetation coverage can reduce soil and water loss effectively and reduce soil total nitrogen losses [19]. The vegetative filter enhances sediment deposition and filtration by vegetation, pollutant adsorption onto soil and plant materials, and uptake of soluble pollutants by plants [20, 21].

A grassed watercourse is a natural or constructed channel with suitable vegetation for safe water disposal. The palatable grass produced from the watercourse is used as fodder for cattle, if it is unpalatable, it can be used as thatching material for huts. Grassed watercourses are also used as filters to remove sediment, but sometimes they may lose their effectiveness when too much sediment builds up in the watercourses. The effectiveness of grassed watercourses in reducing the runoff and sediment loads has been investigated by several researchers [22–27]. Based on the laboratory experiment Briggs et al. reported that grassed watercourses reduced the runoff volume by 47% and herbicide (isoxaben plus oryzalin and isoxaben plus trifluralin) residues by 56% on average comparison to control [24]. Grassed watercourses/terrace systems in potato production with up-and-down slope cultivation practices reduced the average runoff by 86% and the average sediment delivery by 95% [23]. Vegetative watercourses reduce the peak runoff rates, runoff volumes from agricultural watersheds due to their comparably high infiltration rates and the reduction in flow velocity that prolongs the ponding time for infiltration. Various researchers have studied the effectiveness of grassed watercourses in sediment trapping. The flat bottom and V-shaped watercourses reduce the sediment delivery by 97 and 77%, respectively [26]. Fiener and Auerswald [28] reported that the reduction in dissolved reactive phosphorus loads was proportional to the reduction in runoff. The socks installed in the watercourses filtered the surface runoff and reduced the sediment concentration by particle retention within the matrix of the compost and reduced herbicide concentrations due to sorption on the compost [29]. The sediment trapping depends on slope, grass species, soil type, concentration of runoff, and length of the grassed watercourse.

In India, very few studies were conducted to evaluate the grass coverage in watercourses. The effect of grass (Bermuda grass) cover on reduction of runoff, sediment, and flow velocity in watercourses under simulated conditions by using tilting hydraulic flume was reported by several authors [30–32]. The effect of stone and vegetative covers for soil and water conservation in a watercourse on alfisols under simulated conditions by using tilting flume was evaluated [32]. It is also mentioned that these studies did not deal with variable grass coverage [30]. There is a need to study various grass species, their hydraulic roughness, their sediment, and nutrient filtering efficiency under variable grass coverage. In this study, an attempt was made to study the effectiveness of *Dichantium annulatum* as a filter in watercourses, which carried the runoff from agricultural watersheds under various grass coverage for maximum grass production and minimum soil loss.

2 Materials and methods

2.1 Experimental site

This study was conducted during 2012 and 2013 at the Central Soil and Water Conservation Research & Training Institute, Research farm, Vasad (longitude: 73.0806°E, latitude: 22.4574°N), in the Anand District of Gujarat, India. The study area is located in the Mahi River basin of the central part of Gujarat in Western India, 18 km from Anand town and 22 km from Vadodara City of Gujarat. The soil of the experimental site is very deep well drained sandy loam soil. The experimental site soil is poor in fertility and organic carbon ranging from 0.30 to 0.35% and has an infiltration capacity of 3–5 cm/h, a field capacity of 19–20% and a wilting point of 7–8.5%. The last 50 years average weather parameters like annual rainfall at the experimental site is 871 mm and annual maximum and minimum temperatures are 33.7 and 18.9°C, respectively. 94% of the annual rainfall is received in the monsoon season during June to September (Fig. 1).

2.2 Experimental details

A field experiment was conducted to determine the efficacy of *D. annulatum* grass filters in watercourses and to optimize the length of grass coverage in these watercourses. This experiment was conducted with different flow rates (5, 10, 15, and 20 L/s) and with varying grass coverage (0, 25, 50, 75, and 100%) in 2% slope watercourses (Fig. 2). The experiment was carried out with four replications. The watercourses were trapezoidal in shape with a bottom width of 0.6 m, depth of 0.45 m, side slope 1:1, and length of 30 m. For this experiment, *D. annulatum* was used as a vegetative filter in the watercourse. *Dichanthium annulatum* is a natural grass species in this region, which grows well and is very effective in controlling soil loss [33]. *Dichanthium annulatum* not only has a high soil binding ability, survival performance, water stable aggregation, but also has a promising fodder potential [34]. This is one of the best drought-resistant grasses for drought-prone areas in India. *Dichanthium annulatum* was planted as 100% coverage in July, during the monsoon season, and was allowed to grow for 60 days up to the flowering stage in September. The experiment was conducted for the above-mentioned grass coverage cuttings with different discharges. The main treatments include the variations in the length of grass coverage, 100, 75, 50, 25, and 0%. The sub-treatments were variations in discharge 20, 15, 10, and 5 L/s. The rationale behind selecting

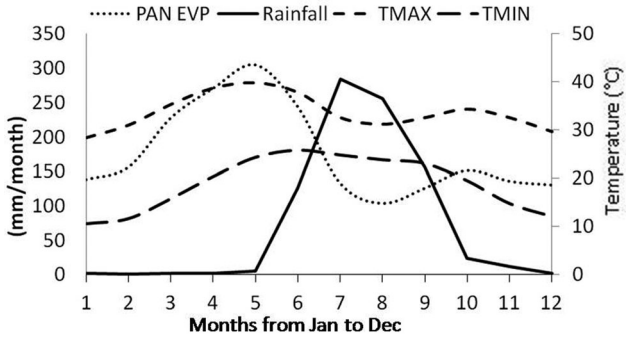


Figure 1. Average of last 50 years weather parameters of experimental site.

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Figure 2. Runoff flow (10 L/s) in vegetative watercourses with varying grass filter coverages, (a) 100%, (b) 50%, (c) 25%, and (d) 0%.



Figure 3. Runoff measurements by H flume and Coshocton wheel sampler.

these discharges was, in a 5 ha watershed, mostly the runoff events, the runoff discharges were in the range of 5–20 L/s. Each discharge was delivered in watercourses for 20 min. To ensure physico-chemical characteristics of real time runoff water passing through the grassed watercourse, proportionate sediments were mixed in the inflow water, which has been pumped from the tube well. Sediment concentration of the inflow with 2 g/L was maintained by manual mixing quantities of sediment in the inflow tank, which has been collected from the silt monitoring structures of the other experiments during the monsoon season. The reason behind for maintaining this sediment concentration was that the average sediment concentration of the experimental data of the various projects in that region is 2 g/L. A similar type of sediment mixing was also reported in the case of a sand-based storm water filtration system for groundwater recharge wells [35]. Before the experiment, the soil bed was brought to field capacity to neutralize the effect of the treatments and to have similar saturated initial conditions.

2.3 Data collection and analysis

The inflow, outflow, flow velocity, depth of the flow, and sediment concentration was measured and analyzed for each run. In one run, one discharge and one coverage were completed. Before the next run, it was always ensured that the soil was brought to field capacity. Finally, 80 runs were generated out of the five main treatments and four sub-treatments with four replications.

2.3.1 Flow measurement

The inflow was measured by a water meter installed in the inflow pipeline. The outflow was measured by means of H-flume (Fig. 3) equipped with an automatic digital stage level recorder. The measured outflow depth or stage was converted into the flow rate by using the stage discharge relationship developed for the given size flume. This flow rate or discharge was converted into the volume unit by multiplying the runoff rate with time.

2.3.2 Sediment concentration

The runoff samples from the outflow were collected by means of Coshocton wheel sampler placed below the H-flume crest for estimating the sediment load (Fig. 3). To estimate the sediment concentration for each setup, water samples of 500 mL at the outflow and inflow for every minute were collected. The outflow runoff samples were collected till the flow recedes. The collected water samples were analyzed using the filtration method after drying in an oven at 105°C.

2.3.3 Flow velocity, depth, Froude number (*F*), Manning's coefficient (*n*)

The flow velocity in a watercourse at various distances was measured by the small pigmy type current meter and it was rechecked by color tracer method and average velocity for each run was estimated. The flow depth in the watercourse at various distances was measured and average depth for each run was calculated. Manning's roughness coefficient (n) and the Froude number (F) were estimated by the following equations.

$$V = \frac{1}{n} R^{2/3} S^{1/2} \tag{1}$$

where V is the flow velocity (m/s), R the hydraulic mean radius (m), and S is the slope.

$$F = \frac{V}{\sqrt{gD}} \tag{2}$$

where D is the hydraulic depth (cross-sectional area of flow/top width) (m) and g is the gravity (m/s^2).

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2.3.4 Statistical analysis

For comparison of the outflow as percentage of the inflow, sediment concentration, V, n, and F, under different coverages and discharges, were analyzed in the split-plot design with considering the percentage of grass coverage as main plots and flow discharges as the sub-plots at 5% significance level.

3 Results and discussion

The results obtained in response to the experiment conducted on the influence of *D. annulatum* grass coverage or grass cutting and flow rate in terms of the outflow, sediment concentration, V, F, and n are discussed below.

3.1 Outflow

The effect of grass coverage on the outflow as percentage of the inflow under various flow rates is presented in Fig. 4. It can be observed that in all cases the outflow as percentage of the inflow is decreased with the increase in grass coverage and a decrease in the flow rate, almost reaching 84% for 100% grass coverage at a lower discharge of 5 L/s. Besides, the outflow increased with the decrease in grass coverage and an increase in the discharge, almost reaching 97% for 0% grass coverage and at the higher discharge of 20 L/s. This trend of decrease in the outflow with increased grass coverage is mainly due to increasing the infiltration opportunity time as a result of resistance of grass coverage to the flow of water. It is also mentioned that an increase in grass coverage might have also increased the deep percolation losses. This type of trend was also reported by the researchers [30, 31]. The statistical analysis for outflow as percentage of inflow under various grass cover and flow rates are presented in Tab. 1. It is observed that among the grass cover treatments, there is no significant difference among the treatments of 100, 75, 50% grass cover. Even among the interactions, there is no significant difference between the treatments of 100, 75, and 50% grass cover under various discharges. This implies that more than 50% filter coverage in watercourses is not much added advantage in reduction of outflow and increase in deep percolation losses. It was also reported that due to the introduction of Bermuda grass in watercourses under alfisols, there is a marginal decrease in outflow [30]. From Tab. 1 it can also be observed that there is significant difference among the flow rate treatments. Moreover, it is observed that for a given grass cover, under lower flow rates; outflow reduction is more pronounced than higher flow rates. This type of trend was also reported by Mishra et al. and Singh et al. [31, 32] who have conducted the experiments under tilting hydraulic flume. This may be due to the fact that under lower discharges, the resistance to flow is higher and thus, increases percolation by dense vegetation. *Dichanthium annulatum* in watercourses was able to reduce the outflow from 12 L/s in 100% grass coverage treatment to 3 L/s in 0% grass coverage treatment when the inflow was 20 L/s. Similarly, when the inflow was 5 L/s, these filters in watercourses were able to reduce the outflow from 16 L/s in 100% grass coverage treatment to 6 L/s in 0% grass coverage treatment.

Table 1. Effect of grass cover and flow discharge on outflow as percentage of inflow

Grass cover (%)	Outflow (%) under various coverages and flow rates				
	20 L/s	15 L/s	10 L/s	5 L/s	Mean
100	87.63	86.25	85.28	83.63	85.7
75	88.28	86.93	85.75	84.33	86.3
50	89.00	87.80	86.60	84.71	87.0
25	92.87	91.39	89.71	86.63	90.2
0	96.90	96.75	94.46	93.66	95.4
Mean	90.94	89.82	88.36	86.59	
		SEM	CD (0.05)		
	GC	0.25	0.98		
	D	0.12	0.36		
	GC × D	0.26	0.80		

SEM, standard error mean; CD, critical difference at 5% significant level; D, discharge, GC, grass cover.

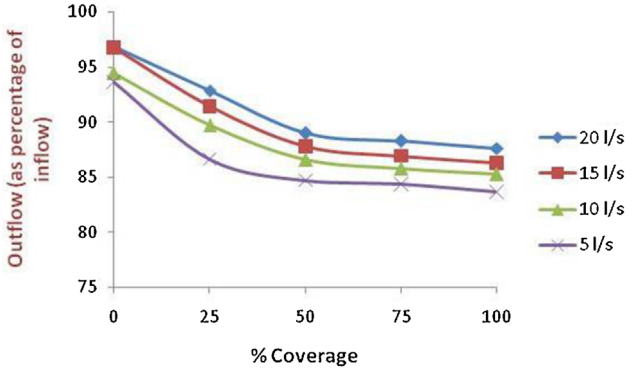


Figure 4. Outflow under various grass coverages and flow rates.

3.2 Sediment concentration

The outflow sediment concentration of a watercourse under various grass coverages and flow rates is presented in Fig. 5. It can be observed that the sediment concentration of the outflow decreases with decreased inflow rates and an increase in grass coverage. In the case of 0% grass coverage with 20 L/s the highest sediment concentration could be observed. The lowest sediment concentration was observed under 100% grass coverage with a flow rate of 5 L/s. The statistical analysis of various grass coverages and flow rates is presented in Tab. 2. It can be observed that there is a significant difference among the flow rate treatments. Moreover, it can also be observed that for a given grass coverage, under lower flow rates, the filtering affect is more pronounced than at higher flow rates. A similar trend was also observed from the experiments conducted under tilting hydraulic flume [31, 32]. *Dichanthium annulatum* in watercourses is able to reduce the sediment concentration in runoff water by 6.5 times (from 4.2 to 0.65 g/L) and 5.7 times (from 2.55 to 0.45 g/L) under higher (20 L/s) and lower (5 L/s) flow rates, respectively. This may be due to the fact that under lower discharges, a higher resistance to flow and a higher filtering capacity at dense vegetation exists. From Tab. 2 it can also be observed that among the grass coverage treatments, there is no significant difference between the treatments of 100, 75, and 50% grass coverage. Even among the interactions, there is no significant difference among the treatments of 100, 75, and 50% grass coverage under various discharges. This implies that at >50% grass coverage in watercourses no additional advantage in reduction of sediment concentration is given. Similar results (25% vegetative cover) were reported by Mishra

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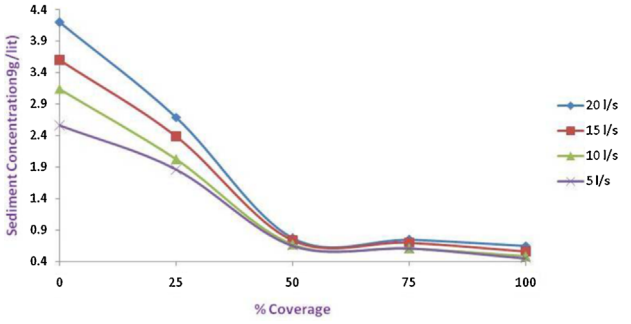


Figure 5. Sediment concentration under various grass coverages and flow rates.

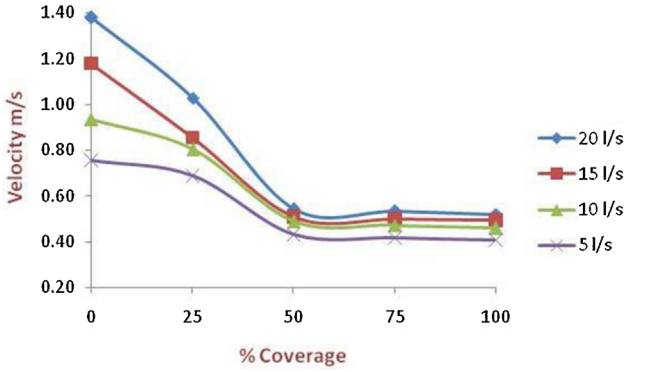


Figure 6. The average velocity of flowing water in a watercourse under various filter coverages and flow rates.

et al. [30] for Bermuda grass in alfisols. Runoff and vegetative ground cover were the main factors that were responsible for the differences in sediment and nitrogen losses [36]. The runoff dynamics and stage of crop were the main responsible factors for sediment and nitrogen losses from agricultural watersheds [37]. It can also be observed that for <50% grass coverage treatment, irrespective of the discharge, erosion started. The outflow sediment concentration was higher than the inflow sediment concentration for treatments with <50% grass coverage and it is vice versa for treatments with ≥50% grass coverage. This may be due to the fact that the flow velocity was almost non-erosive up to 50% grass coverage. The decrease in the sediment concentration of the outflow with an increase in grass coverage is mainly due to decrease in velocity thereby decreasing the transport capacity, filtering of sediment by grass coverage. Based on these results, it can be inferred that in grassed watercourses 50% grass coverage can be harvested as fodder without much compromising on sediment loss.

3.3 Flow velocity

The average velocity of flowing water in a watercourse under various filter coverages and discharges is presented in Fig. 6. From Fig. 6 it can be observed that the velocity decreased with an increase in grass coverage and a decrease in discharge. At a lower coverage, the velocity is higher and flows were super-critical and critical, only when the grass coverage increased by up to 50%, the velocity is drastically reduced. Beyond this, the velocity is reduced gradually with an increase in grass coverage. The statistical analysis for velocity under various grass coverages and flow rates is presented in Tab. 3. From Tab. 3 it can be observed that among the grass coverage treatments, there is no significant difference for 100, 75, and 50%. Even among the interactions, there is no significant difference between the 100, 75, and 50% grass coverage treatments under various discharges. This implies that for >50% filter coverage in watercourses no additional advantage in reduction of velocity and thus, a decrease in soil erosion. There is not much added advantage of 100% of Bermuda grass cover, >75% cover for reduction of flow velocity in watercourses under alfisols [30]. From Tab. 3 it can also be observed that there is a significant difference among the flow rate treatments. Moreover, it can be observed that for a given grass coverage, under lower flow rates, the velocity reduction is more pronounced than for higher flow rates. This may be due that under lower discharges, more resistance to flow and. This may be due to the fact that under lower discharges, a higher resistance to flow exists and thus, a decrease of flow velocity for dense vegetation can be observed. *Dichanthium annulatum* in watercourses is able to reduce the flow velocity of runoff water by two times (from 1.06 to 0.47 m/s).

Table 2. Effect of grass cover and flow discharge on sediment concentration

	Sediment concentration (g/L) under various coverages and flow rates				
Grass cover (%)	20 L/s	15 L/s	10 L/s	5 L/s	Mean
100	0.65	0.56	0.49	0.45	0.54
75	0.74	0.70	0.59	0.61	0.66
50	0.77	0.74	0.67	0.65	0.71
25	2.69	2.39	2.02	1.86	2.24
0	4.20	3.60	3.14	2.56	3.37
Mean	1.81	1.60	1.39	1.22	
		SEM	CD (0.05)		
	GC	0.08	0.32		
	D	0.02	0.05		
	GC × D	0.04	0.12		

SEM, standard error mean; CD, critical difference at 5% significant level; D, discharge; GC, grass cover.

Table 3. Effect of grass cover and flow rate on flow velocity

	Flow velocity (m/s) under various coverages and flow rates				
Grass cover (%)	20 L/s	15 L/s	10 L/s	5 L/s	Mean
100	0.521	0.493	0.460	0.408	0.471
75	0.533	0.498	0.473	0.418	0.480
50	0.543	0.508	0.488	0.433	0.493
25	1.029	0.855	0.802	0.689	0.844
0	1.381	1.180	0.933	0.755	1.062
Mean	0.801	0.707	0.631	0.540	
		SEM	CD (0.05)		
	GC	0.004	0.015		
	D	0.004	0.012		
	GC × D	0.009	0.027		

SEM, standard error mean; CD, critical difference at 5% significant level; D, discharge; GC, grass cover.

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3.4 Manning's roughness coefficient (*n*) and Froude number (*F*)

Manning's roughness coefficient (*n*) of a grassed watercourse under various grass filter coverages and flow rates is presented in Tab. 4. It can be observed that *n* is significantly different among the flow rate treatments. Moreover, it can be observed that for a given grass coverage, under lower flow rates, the roughness effect is more pronounced than for higher flow rates. This indicates that vegetative coverage offers more resistance to the flow. The resistance has diminished with the increase in inflow discharge. From Tab. 4 it can also be observed that *n* is not significantly different among the grass coverage treatments of 100, 75, and 50% under various flow rates. Same was also noticed among the interactions. This implies that for >50% filter coverage in watercourses there is no additional advantage in the increase in the roughness of channels. It was also reported that *n* could be increased by three times for 100% vegetation coverage [38]. The *n* values were more for cropped furrow conditions than for bare furrow [39]. The filter strips are able to increase *n* from 0.0323 to 0.1006 under various discharges. The statistical analysis of *F* of grassed watercourses under various grass filter coverages and flow rates is presented in Tab. 5. From Tab. 5 it can be observed that *F* not statistically differs among grass coverage treatments of 100, 75, 50% under various flow rates and the same is also noticed among the interactions, but is significantly different among the flow rate treatments. From Tab. 5 it can be observed that the grass coverage treatments convert the super-critical flows into sub-critical flows, thereby reducing the scouring and sediment transport.

Table 4. Effect of grass cover and flow rate on Manning's roughness coefficient (*n*)

	n under various coverages and flow rates				
Grass cover (%)	20 L/s	15 L/s	10 L/s	5 L/s	Mean
100	0.095	0.097	0.101	0.109	0.101
75	0.090	0.099	0.103	0.105	0.099
50	0.081	0.091	0.097	0.104	0.093
25	0.043	0.052	0.054	0.060	0.052
0	0.026	0.029	0.035	0.040	0.032
Mean	0.067	0.073	0.078	0.083	
		SEM	CD (0.05)		
	GC	0.0009	0.0035		
	D	0.0006	0.0018		
	GC × D	0.0013	0.0240		

SEM, standard error mean; CD, critical difference at 5% significant level; D, discharge; GC, grass cover.

Table 5. Effect of grass cover and flow rate on Froude number (*F*)

	F under various flow rates				
Grass cover (%)	20 L/s	15 L/s	10 L/s	5 L/s	Mean
100	0.366	0.355	0.337	0.310	0.34
75	0.38	0.35	0.33	0.32	0.35
50	0.40	0.35	0.33	0.31	0.35
25	0.75	0.63	0.59	0.53	0.62
0	1.25	1.06	0.86	0.75	0.98
Mean	0.63	0.55	0.49	0.44	
		SEM	CD (0.05)		
	GC	0.01	0.04		
	D	0.00	0.01		
	GC × D	0.01	0.02		

SEM, standard error mean; CD, critical difference at 5% significant level; D, discharge; GC, grass cover.

4 Concluding remarks

The grass filter strips are able to reduce the sediment concentration in runoff water by six times (from 4.2 to 0.65 g/L) and they are able to reduce the velocity of runoff water by two times (from 1.06 to 0.47 m/s) and convert the super-critical flows into sub-critical flows. The filter strips are able to increase *n* from 0.0323 to 0.1006 under various discharges. It was also found that for >50% filter coverage in watercourses no additional advantage in reduction of sediment concentration, outflow, flow velocity, and reduction of *F*, and increase in roughness of the channels exists. Based on these results, *D. annulatum* can be used as a filter in watercourses to prevent runoff and soil losses from agricultural watersheds and thereby reducing the sedimentation of downstream water bodies. The optimum grass coverage, i.e. 50% of the channel length, can maintain the maximum grass production with minimum soil loss.

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References

[1] M. G. Dosskey, Toward Quantifying Water Pollution Abatement in Response to Installing Buffers on Cropland, *Environ. Manage.* **2001**, *28*, 577–598.

[2] M. S. Rama Mohan Rao, R. N. Adhikari, S. Chittaranjan, M. Chandrappa, M. Padmaiah, Influence of Conservation Measures on Water Yield from Catchments in the Semi-arid Tract of South India, *J. Hydrol.* **1993**, *31* (*1*), 23–38.

[3] A. K. Singh, M. S. Rama, R. Mohan, K. Batchelor, Mukherjee, Impact of Watershed Development on Traditional Tank Systems, *J. Rural Dev.* **2004**, *23* (*1*), 59–72.

[4] K. T. Krishnagowda, A. M. Krishnappa, P. Panduranga, B. R. Hegde, Live bunds for soil and moisture conservation under dryland conditions, *International Symposium on Water Erosion, Sedimentation and Resource Conservation*, October 9–13, CSWCRTI, Dehradun **1990**.

[5] D. H. Ranade, R. A. Sharma, R. K. Gupta, A. N. Patel, Effect of Mechanical and Vegetative Barriers on Conservation of Runoff, Soil and Plant Nutrients, *Crop Res.* **1995**, *9* (*2*), 218–223.

[6] C. Prakash, A. Raizada, J. S. Samra, G. Sastry, *Vegetative Barriers for Resource Conservation*, Technical Bulletin No. T-41/D-29, Central Soil and Water Conservation Research and Training Institute (ICAR), Dehradun, Uttaranchal, India **1999**.

[7] S. N. Prasad, R. K. Singh, S. Ali, A. K. Prandiyal, Comparative Performance of Grass Barriers on Erosion and Crop Yields in Medium Black Soils of Kota, *Indian J. Soil Conserv.* **2005**, *33* (*1*), 58–56.

[8] A. Dass, U. S. Patnaik, S. Sudhishri, N. K. Paikaray, V. K. Dwivedi, Performance of Lemongrass (*Cymbopogon citratus*) as Vegetative Barrier in Eastern Ghats of Orissa, *Indian For.* **2006**, *13* (*9*), 1189–1194.

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[9] S. C. Senapati, S. D. Sharma, Effect of Buffer Strips on Soil Erosion and Yield of Maize (*Zea mays* L.) in Rainfed Sloppy Uplands in North-western Plateau Zone of Orissa, *Indian J. Soil Conserv.* **2007**, *35* (*1*), 47–49.

[10] S. Sudhishri, A. Dass, N. K. Lenka, Efficacy of Vegetative Barriers for Rehabilitation of Degraded Hill Slopes in Eastern India, *Soil Tillage Res.* **2008**, *99*, 98–107.

[11] A. Dass, S. Sudhishri, N. K. Lenka, U. S. Patnaik, Runoff Capture through Vegetative Barriers and Planting Methodologies to Reduce Erosion, and Improve Soil Moisture, Fertility and Crop Productivity in Southern Orissa, India, *Nutr. Cycl. Agroecosyst.* **2011**, *89*, 45–57.

[12] T. A. Dillaha, R. B. Reneau, S. Mostaghimi, D. Lee, Vegetative Filter Strips for Agricultural Nonpoint Source Pollution Control, *Trans. ASAE* **1989**, *32*, 513–519.

[13] L. Patty, B. Real, J. J. Gril, The Use of Grassed Buffer Strips to Remove Pesticides, Nitrate and Soluble Phosphorus Compounds, *Pestic. Sci.* **1997**, *49*, 243–251.

[14] K. Arora, S. K. Mickelson, J. L. Baker, D. P. Tierney, C. J. Peters, Herbicide Retention by Vegetative Buffer Strips from Runoff under Natural Rainfall, *Trans. ASAE* **1996**, *39*, 2155–2162.

[15] H. Ghadiri, C. W. Rose, W. L. Hogarth, The Influence of Grass and Porous Barrier Strips on Runoff Hydrology and Sediment Transport, *Trans. ASAE* **2001**, *44*, 259–268.

[16] H. Blanco-Canqul, C. J. Gantzes, S. H. Anderson, E. E. Alberto, A. L. Thompson, Grass Barrier and Vegetative Filterstrip Effectiveness in Reducing Runoff, Sediment, Nitrogen and Phosphorus Loss, *Soil Sci. Soc. Am. J.* **2004**, *68*, 1670–1678.

[17] M. Duchemin, R. Hogue, Reduction in Agricultural Non-Point Source Pollution in the First Year Following Establishment of an Integrated Grass/Tree Filter Strip System in Southern Quebec (Canada), *Agric. Ecosyst. Environ.* **2009**, *131* (*1–2*), 85–97.

[18] M. Borin, M. Vianello, F. Morari, G. Zanin, Effectiveness of Buffer Strips in Removing Pollutants in Runoff from a Cultivated Field in North-East Italy, *Agric. Ecosyst. Environ.* **2005**, *105*, 101–114.

[19] X. C. Zhang, G. B. Liu, H. F. Fu, Soil Nitrogen Losses of Catchment by Water Erosion as Affected by Vegetation Coverage, *Environ. Sci.* **2000**, *21* (*6*), 16–19.

[20] A. K. Misra, J. L. Baker, S. K. Mickelson, H. Shang, Contributing Area and Concentration Effects on Herbicide Removal by Vegetated Bufferstrips, *Trans. ASAE* **1996**, *39*, 2105–2111.

[21] S. B. Blanche, D. R. Shaw, J. H. Massey, M. Boyette, M. C. Smith, Fluometuron Adsorption to Vegetative Filter Strip Components, *Weed Sci.* **2003**, *51*, 125–129.

[22] A. Hjelmfelt, M. Wang, Modeling Hydrologic and Water Quality Responses to Grass Waterways, *J. Hydrol. Eng.* **1999**, *4*, 251–256.

[23] T. L. Chow, H. W. Rees, J. L. Daigle, Effectiveness of Terraces/Grassed Waterway Systems for Soil and Water Conservation: A Field Evaluation, *J. Soil Water Conserv.* **1999**, *54* (*3*), 577–583.

[24] J. A. Briggs, T. Whitwell, M. B. Riley, Remediation of Herbicides in Runoff Water from Container Plant Nurseries Utilizing Grassed Water Ways, *Weed Technol.* **1999**, *12*, 157–164.

[25] P. Fiener, K. Auerswald, Concept and Effects of a Multi-purpose Grassed Waterway, *Soil Use Manage.* **2003**, *19*, 65–72.

[26] P. Fiener, K. Auerswald, Effectiveness of Grassed Waterways in Reducing Runoff and Sediment Delivery from Agricultural Watersheds, *J. Environ. Qual.* **2003**, *30*, 927–936.

[27] P. Fiener, K. Auerswald, Measurement and Modeling of Concentrated Runoff in a Grassed Waterway, *J. Hydrol.* **2005**, *301*, 198–215.

[28] P. Fiener, K. Auerswald, Effects of Hydrodynamically Rough Grassed Waterways on Dissolved Reactive Phosphorus Loads Coming from Agricultural Watersheds, *J. Environ. Qual.* **2009**, *38*, 548–559.

[29] J. M. Shipitalo, J. V. Bonta, E. A. Dayton, L. B. Owens, Impact of Grassed Waterways and Compost Filter Socks on the Quality of Surface Runoff from Corn Fields, *J. Environ. Qual.* **2010**, *39* (*3*), 1009–1018.

[30] P. K. Mishra, J. K. Neelkanth, B. Maheswara Babu, S. S. Kumathe, Effectiveness of Bermuda Grass as Vegetative Cover in Grassed Water Way: A Simulated Study, *J. Irrigat. Drain. Eng.* **2006**, *132* (*3*), 288–292.

[31] A. K. Singh, M. Kothari, V. Kumar, R. N. Adhikari, M. Singh, Development of Tilting Hydraulic Flume for Runoff and Soil Loss Studies in Laboratory, *Indian J. Soil. Conserv.* **2010**, *38* (*2*), 86–88.

[32] B. Narsimlu, B. Venkateswara Rao, P. K. Mishra, K. V. Rao, Effect of Covers for Soil and Water Conservation Using Tilting Flume, *J. Irrigat. Drain. Eng.* **2004**, *47*, 154–159.

[33] S. P. Tiwari, R. S. Kurothe, Effect of Vegetative Barriers on Soil and Nutrients Losses at 2% Slope on Agricultural Lands of Reclaimed Mahi Ravines, *Indian J. Soil Conserv.* **2006**, *34* (*1*), 37–41.

[34] G. L. Chunale, Evaluation of Different Grass Species for Soil Binding and Soil Aggregation Properties under Submontane Zone of Maharashtra, *Indian J. Soil Conserv.* **2004**, *32* (*1*), 24–27.

[35] S. Kumar, S. K. Kamra, R. K. Yadav, J. P. Sharma, Evaluation of Sand-based Stormwater Filtration System for Groundwater Recharge Wells, *Curr. Sci.* **2012**, *103* (*4*), 395–404.

[36] R. Vasquez-Mendez, E. Ventura-Ramos, K. Oleschko, L. Hernandez-Sandoval, J.-F. Parrot, M. A. Neraring, Soil Erosion and Runoff in Different Vegetation Patches from Semiarid Central Mexico, *Catena* **2010**, *80*, 162–169.

[37] Q. Liu, Z. Li, P. Li, Effect of Runoff Dynamic on Sediment and Nitrogen Losses in an Agricultural Watershed of the Southern Shaanxi Region, China, *Clean – Soil Air Water* **2014**, *42* (*1*), 56–63.

[38] A. R. Sepaskhah, H. Bondar, Estimation of Manning Roughness Coefficient for Bare and Vegetated Furrow Irrigation, *Biosyst. Eng.* **2002**, *82* (*3*), 351–357.

[39] D. Mailapalli, N. Raghuwanshi, R. Singh, G. Schmitz, F. Lennartz, Spatial and Temporal Variations of Manning's Roughness Coefficient in Furrow Irrigation, *J. Irrigat. Drain. Eng.* **2008**, *134* (*2*), 185–192.

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